

Fall 1999

Problem 1 (Fa99). Let V and W be finite dimensional vector spaces, let X be a subspace of W , and let $T : V \rightarrow W$ be a linear map. Prove that the dimension of $T^{-1}(X)$ is at least $\dim V - \dim W + \dim X$.

Problem 2 (Fa99). Let $E_1, E_2, \dots$ be a sequence of nonempty closed subsets of a complete metric space (X, d) with $E_{n+1} \subset E_n$ for all positive integers n , and such that $\lim_{n \rightarrow \infty} \text{diam}(E_n) = 0$, where $\text{diam}(E)$ is defined to be

$$\sup\{d(x, y) \mid x, y \in E\}$$

Prove that $\bigcap_{n=1}^{\infty} E_n \neq \emptyset$.

Problem 3 (Fa99). Let R be a ring with identity. Suppose that $\mathfrak{I}_1, \mathfrak{I}_2, \dots, \mathfrak{I}_n$ are left ideals in R such that $R = \mathfrak{I}_1 \oplus \mathfrak{I}_2 \oplus \dots \oplus \mathfrak{I}_n$ (as additive groups). Prove that there are elements $u_i \in \mathfrak{I}_i$ such that for any elements $a_i \in \mathfrak{I}_i$, $a_i u_i = a_i$ and $a_i u_j = 0$ if $j \neq i$.

Problem 4 (Fa99). Let the rational function f in the complex plane have no poles for $\Im z \geq 0$. Prove that

$$\sup\{|f(z)| \mid \Im z \geq 0\} = \sup\{|f(z)| \mid \Im z = 0\}$$

Problem 5 (Su78, Fa99). Let $M_{n \times n}$ denote the vector space of real $n \times n$ matrices. Define a map $f : M_{n \times n} \rightarrow M_{n \times n}$ by $f(X) = X^2$. Find the derivative of f .

Problem 6 (Fa99). Let $T : V \rightarrow V$ be a linear operator on an n dimensional vector space V over a field $\mathbb{F}$. Prove that T has an invariant subspace W other than $\{0\}$ and V if and only if the characteristic polynomial of T has a factor $f \in \mathbb{F}[t]$ with $0 < \deg f < n$.

Problem 7 (Fa99, Sp14). Let G be a finite group acting on a finite set X with a single orbit. For an element $g \in G$ let $\text{Fix}_g(X)$ denote the set $\{x \in X \mid g(x) = x\}$.

1. Show that $|G| = \sum_{g \in G} |\text{Fix}_g(X)|$

2. Show that if X has more than one point then there exists an element $g \in G$ fixing no points of X .

Problem 8 (Fa99, Sp12). Evaluate the integral

$$I = \frac{1}{2\pi i} \int_{|z|=1} \frac{(z+2)^2}{z^2(2z-1)} dz$$

where the direction of integration is counterclockwise.

Problem 9 (Fa99). Describe all three dimensional vector spaces V of $\mathcal{C}^\infty$ complex valued functions on $\mathbb{R}$ that are invariant under the operator of differentiation.

Problem 10 (Fa99, Sp24). Let f be a continuous real valued function on $[0, \infty)$ such that $\lim_{x \rightarrow \infty} f(x)$ exists (finitely). Prove that f is uniformly continuous.

Problem 11 (Fa83, Sp87, Fa99). Let V be a finite dimensional complex vector space and let A and B be linear operators on V such that $AB = BA$. Prove that if A and B can each be diagonalized, then there is a basis for V which simultaneously diagonalizes A and B .

Problem 12 (Fa99). Let $A = \{0\} \cup \{1/n \mid n \in \mathbb{Z}, n > 1\}$ and let $\mathbb{D}$ be the open unit disc in the complex plane. Prove that every bounded holomorphic function on $\mathbb{D} \setminus A$ extends to a holomorphic function on $\mathbb{D}$.

Problem 13 (Fa99). Let $\mathbb{K}$ be the field $\mathbb{Q}(\sqrt[10]{2})$. Prove that $\mathbb{K}$ has degree 10 over $\mathbb{Q}$, and that the group of automorphisms of $\mathbb{K}$ has order 2.

Problem 14 (Sp92, Fa99). Show that every infinite closed subset of $\mathbb{R}^n$ is the closure of a countable set.

Problem 15 (Fa99). Let A be an $n \times n$ complex matrix such that $\text{tr } A^k = 0$ for $k = 1, \dots, n$. Prove that A is nilpotent.

Problem 16 (Fa99). For $0 < a < b$, evaluate the integral

$$I = \frac{1}{2\pi} \int_0^{2\pi} \frac{1}{|ae^{i\theta} - b|^4} d\theta$$

Problem 17 (Fa99). Show that a group G is isomorphic to a subgroup of the additive group of the rationals if and only if G is countable and every finite subset of G is contained in an infinite cyclic subgroup of G .

Problem 18 (Fa99). Let f and g be continuous real valued functions on $\mathbb{R}$ such that $\lim_{|x| \rightarrow \infty} f(x) = 0$ and $\int_{-\infty}^{\infty} |g(x)| \, dx < \infty$. Define the function h on $\mathbb{R}$ by

$$h(x) = \int_{-\infty}^{\infty} f(x-y)g(y) \, dy$$

Prove that $\lim_{|x| \rightarrow \infty} h(x) = 0$.